Lect-1: Probability Sec. B

Fermat, Pascal, Newton

Statistics is the logic of uncertainty

Sample space: The set of all possible outcomes of an experiment.

<u>Event</u> is a subset of the sample space.

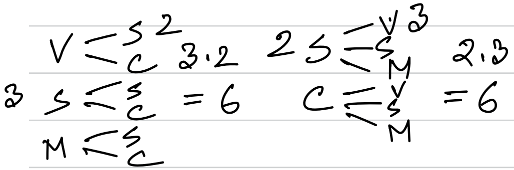
Naive def^n of probability: $P(\overset{\to \text{event}}{A}) = \dfrac{\#\text{favorable outcomes}}{\#\text{total outcomes}}$

Flip a coin twice: HH, HT, TH, TT

Event A: At least one H. $P(A) = 3/4$

Assumes all outcomes are equally likely & finite sample space

Counting: Multiplication rule: If one experiment has n_1 outcomes and for each outcome of the 1st experiment, there are n_2 possible outcomes for the 2nd experiment .... n_r possible outcomes for the r^{th} experiment, then there are $n_1 n_2 \cdots n_r$ overall possible outcomes.



52 cards, 4 suites, 13 kinds in each suite

Ex. Probability of full house in poker with a 5-card hand. Three cards of one kind, two cards of another.

$$P(\overset{\to \text{full house}}{A}) = \frac{13 \cdot \binom{4}{3} \cdot 12 \cdot \binom{4}{2}}{\binom{52}{5}}$$

Sampling table: Choose k objects out of n.

	Order (Y)	Order (N)
Replace (V)	n^k	$\binom{n+k-1}{k}$
No replace (X)	$n!/(n-k)!$	$\binom{n}{k} = \dfrac{n!}{(n-k)!\,k!}$

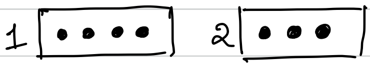
$$n \cdot (n-1) \cdot (n-2) \cdots (n-k+1) = \frac{n \cdot (n-1) \cdots 2 \cdot 1}{1 \cdot 2 \cdots (n-k)} = n!/(n-k)! = {}^nP_k$$

Lect 2: Counting & Story Proofs Sec. B

Choose k objects out of n; sample k times from n objects where order does not matter and replacement is allowed

$\binom{n+k-1}{k}$; $k=0 \Rightarrow \binom{n-1}{0}=1$, $k=1 \Rightarrow \binom{n}{1}=n$, $n=1 \Rightarrow \binom{k}{k}=1$

$n=2 \Rightarrow \binom{k+1}{k}=k+1$



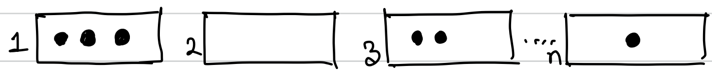
1122

1221

11221211

Equiv: How many ways to put k indistinguishable objects into n distinguishable boxes?

k dots, $n-1$ seps



$\Rightarrow$ ••• || •• | .... | •

$\frac{(n+k-1)!}{k!(n-1)!} = \binom{n+k-1}{k}$

Story proofs: (1) $\binom{n}{k} = \binom{n}{n-k}$

(2) $n\binom{n-1}{k-1} = k\binom{n}{k}$ [Pick k people out of n, designate a captain]

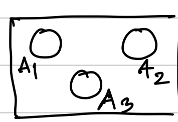
(3) $\binom{m+n}{k} = \sum_{j=0}^{k} \binom{m}{j}\binom{n}{k-j}$ [Vandermonde's identity]

Non-naive def^n: A probability space consists of S and P, where S is a sample space and P is a function which takes an event $A \subseteq S$ as input and gives $P(A) \in [0,1]$ such that

(1) $P(\emptyset)=0$, $P(S)=1$

(2) $P\left(\bigcup_{k=1}^{\infty} A_k\right) = \sum_{k=1}^{\infty} P(A_k)$, if A_k are disjoint

S



Lect 8: Birthday Problem Sec. B

k people, find the probability that two of them have same birthdays.
If $k > 365$, then $P=1$ (from pigeonhole principle)
Equally likely birthdays, assume independent birthdays.

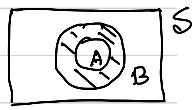
Let $k \le 365$, $P(\text{no match}) = \dfrac{365 \cdot 364 \cdot \dots \cdot (365-k+1)}{365^k}$ $\quad \binom{23}{2} = 253$

$$P(\text{match}) = 1 - P(\text{no match}) = \begin{cases} 50.7\%, & k=23 \\ 97\%, & k=50 \\ 99.999\%, & k=100 \end{cases}$$

Properties of probability: (1) $P(A^c) = 1 - P(A)$

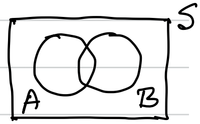
$1 = P(S) = P(A \cup A^c) = P(A) + P(A^c) \Rightarrow P(A^c) = 1 - P(A)$

(2) If $A \subseteq B$, then $P(A) \le P(B)$



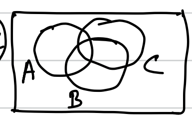
Proof: $B = A \cup (B \cap A^c) \Rightarrow P(B) = P(A) + P(B \cap A^c) \ge P(A)$

(3) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$



Proof: $P(A \cup B) = P(A \cup (B \cap A^c)) = P(A) + P(B \cap A^c)$
Now, we need to prove $P(B) - P(A \cap B) = P(B \cap A^c)$
$\Leftrightarrow P(B) = P(B \cap A) + P(B \cap A^c)$ which is true [Disjoint union]

$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C)$
Principle of inclusion and exclusion. $+ P(A \cap B \cap C)$



$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i) - \sum_{i<j} P(A_i \cap A_j) + \sum_{i<j<k} P(A_i \cap A_j \cap A_k) + \dots + (-1)^{n+1} P\left(\bigcap_{i=1}^n A_i\right)$$

deMontmort's matching problem:

3	1	5	4	2
1	2	3	4	5

$P(\bigcup_{i=1}^n A_i)$; A_j be the event that the j^{th} element matches

$$P(A_j) = \frac{1 \cdot (n-1) \cdot \dots \cdot 2 \cdot 1}{n \cdot (n-1) \cdot \dots \cdot 2 \cdot 1} = \frac{(n-1)!}{n!} = \frac{1}{n}; \quad P(A_i \cap A_j) = \frac{1 \cdot 1 \cdot (n-2) \cdot \dots \cdot 2 \cdot 1}{n \cdot (n-1) \cdot \dots \cdot 2 \cdot 1} = \frac{(n-2)!}{n!} = \frac{1}{n(n-1)}$$

$$P(A_1 \cap A_2 \cdots \cap A_k) = \frac{1 \cdots 1 \cdot (n-k) \cdots 2 \cdot 1}{n(n-1) \cdots 2 \cdot 1} = \frac{(n-k)!}{n!}$$

Lect. 4: Sec. B

$$P(\cup_{i=1}^{n} A_i) = \sum_{i=1}^{n} P(A_i) - \sum_{i<j} P(A_i \cap A_j) + \sum_{i<j<k} P(A_i \cap A_j \cap A_k) + \cdots + (-1)^{n+1} P(\cap_{i=1}^{n} A_i)$$

$$P(A_j) = \frac{1}{n} \; ; \; P(A_i \cap A_j) = \frac{1}{n(n-1)} \; ; \; P(A_1 \cap A_2 \cap \cdots \cap A_k) = \frac{(n-k)!}{n!}$$

$$P(\cup_{i=1}^{n} A_i) = n \cdot \frac{1}{n} - \frac{n \cdot (n-1)}{2} \times \frac{1}{n(n-1)} + \cdots \; \frac{n!}{(n-k)!\,k!} \times \frac{(n-k)!}{n!} \cdots + (-1)^{n+1} \frac{1}{n!}$$

$$= 1 - \tfrac{1}{2} + \cdots + (-1)^{k+1} \tfrac{1}{k!} - \cdots (-1)^{n+1} \frac{1}{n!}$$

$$= \tfrac{1}{1!} - \tfrac{1}{2!} + \tfrac{1}{3!} - \cdots (-1)^{n+1} \tfrac{1}{n!} \qquad [e^{-1} = 1 - \tfrac{1}{1!} + \tfrac{1}{2!} - \tfrac{1}{3!} + \cdots]$$

$$\approx 1 - e^{-1} \qquad \Rightarrow 1 - e^{-1} = \tfrac{1}{1!} - \tfrac{1}{2!} + \tfrac{1}{3!} - \cdots$$

$$P(A_1^c \cap A_2^c \cdots \cap A_n^c) \approx e^{-1} = 0.367$$

Conditional Probability:

Def^n: Event A, B are independent if $P(A \cap B) = P(A)P(B)$

A, B, C are independent if $P(A, B, C) = P(A)P(B)P(C)$

$P(A \cap B) = P(A)P(B)$, $P(B \cap C) = P(B)P(C)$; $P(A \cap C) = P(A)P(C)$

Newton-Pepys Problem (1697): Have fair dice, which is more likely?

(A) at least 1 six with 6 dice; $P(A) = 1 - \frac{5}{6} \cdot \frac{5}{6} \cdots \frac{5}{6} = 1 - (\frac{5}{6})^6 = 0.665$

(B) " " 2 sixes " 12 "; $P(B) = 1 - (\frac{5}{6})^{12} - 12 \cdot \frac{1}{6} \cdot (\frac{5}{6})^{11} = 0.619$

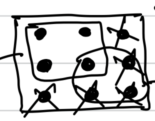
(C) " " 3 " " 18 "; $P(C) = 1 - \sum_{k=0}^{2} \binom{18}{k} (\frac{1}{6})^k (\frac{5}{6})^{18-k} = 0.597$

Conditional Probability: How should we update prob/belief/uncertainty when new information is revealed to us!

Def^n: $P(A|B) = P(A \cap B) / P(B)$, $P(B) > 0$

S

B



A

9 dots, equally likely. $P(A \cap B) = \frac{1}{9}$ $\quad P(B) = \frac{4}{9}$

$P(A|B) = \frac{1}{4}$

$\frac{1}{9} \div x = \frac{1}{4} \Rightarrow x = \frac{4}{9}$

$P(A \cap B) / P(B) = P(A|B)$

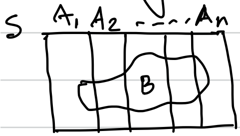
Def^n: $P(A|B) = P(A\cap B)/P(B)$, $P(B)>0$

Th^{m}1: $P(A\cap B) = P(B)P(A|B) = P(A)P(B|A)$

Th^{m}2: $P(A_1\cap A_2\cap \cdots \cap A_n) = P(A_1)P(A_2|A_1)\cdots P(A_n|A_1A_2\cdots A_{n-1})$

Th^{m}3: $P(A|B) = P(A)P(B|A)/P(B)$ [Bayes' rule]

Thinking conditionally, how to solve problems.



Let $A_1, A_2, \ldots, A_n$ be a partition of S.

$$P(B) = P(B\cap A_1) + P(B\cap A_2) + \cdots + P(B\cap A_n)$$
$$= P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + \cdots + P(A_n)P(B|A_n)$$

→ Law of total probability

Ex. Get random 2-card hand from a standard deck.

$P(\text{both aces} \mid \text{have ace})$

$$= \frac{P(\text{both aces, have ace})}{P(\text{have ace})} = \frac{\binom{4}{2}/\binom{52}{2}}{1-\binom{48}{2}/\binom{52}{2}} = \frac{1}{33} = 3.03\%$$

$P(\text{both aces} \mid \text{have an ace of spades})$

$$= \frac{P(\text{both aces, have ace spades})}{P(\text{Ace of spades})} = 3/51 = 1/17 = 5.88\%$$

$\boxed{AS}$ $\boxed{}$

Disease affects 1% population. $P(D)=0.01$ D: patient has disease

Suppose test is 95% accurates. $P(D^c)=0.99$ T: tests positive

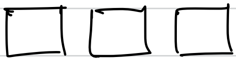
$P(T|D) = 0.95 = P(T^c|D^c)$

$P(T^c|D) = 0.05 = P(T|D^c)$

$$P(D|T) = \frac{P(D\cap T)}{P(T)} = \frac{P(D\cap T)}{P(T\cap D)+P(T\cap D^c)}$$
$$= \frac{P(D)P(T|D)}{P(D)P(T|D)+P(D^c)P(T|D^c)}$$
$$= \frac{0.95\times 0.01}{0.95\times 0.01 + 0.99\times 0.05} = 0.16$$

Def^n: Events A, B are independent conditionally given C if $P(A\cap B|C) = P(A|C)P(B|C)$

Lect. 6: Monty Hall Sec B



1 door has car, 2 has goats. Monty knows which is which. After picking a door, Monty always opens a goat door. If he has a choice which door to open, he opens with equal probability. Should we switch?

Choice — Car door — Monty opens

1 $\xrightarrow{1/3}$ 1 $\xrightarrow{1/2}$ 2

1 $\xrightarrow{1/3}$ 1 $\xrightarrow{1/2}$ 3

1 $\xrightarrow{1/3}$ 2 $\xrightarrow{1}$ 3

1 $\xrightarrow{1/3}$ 3 $\xrightarrow{1}$ 2

$1/6 \rightarrow \frac{1/6}{1/6+1/3} = \frac{1}{3}$

$1/3 \rightarrow \frac{1/3}{1/6+1/3} = 2/3$

S: success (if we switch)

D_j: Door j has car

$j \in \{1,2,3\}$

$P(S)$

P(success of switching | Monty opens 2) $= 2/3$

$P(S) = P(S\cap D_1) + P(S\cap D_2) + P(S \cap D_3)$

$\quad = P(D_1)P(S|D_1) + P(D_2)P(S|D_2) + P(D_3)P(S|D_3)$

$\quad = 1/3 \cdot 0 + 1/3 \cdot 1 + 1/3 \cdot 1 = 2/3$

Simpson's Paradox:

Hibbert

	heart	bandage	
S	70	10	80%
F	20	0	
	77%	100%	

Nick

	heart	bandage	
S	0	81	81%
F	10	9	
	0%	90%	

A: Successful surgery $\quad P(A|B,C) < P(A|B^c,C)$

B: treated by Dr. Nick $\quad P(A|B,C^c) < P(A|B^c,C^c)$

C: heart surgery $\quad$ but, $P(A|B) > P(A|B^c)$

$P(A|B) = P(A \cap C|B) + P(A\cap C^c|B)$

$\quad = P(C|B)\underbrace{P(A|B,C)}_{<P(A|B^c,C)} + P(C^c|B)\underbrace{P(A|B,C^c)}_{<P(A|B^c,C^c)}$

Lect. 7: Gambler's Ruin Sec. B

Two gamblers, A & B. Sequence of rounds, bet \$1.
Let P(A wins round) = p & $q = 1-p$. Find probability that A wins game (B gets bankrupt) assuming A starts with \$$i$ and B with \$$(N-i)$

Random walk $0 \quad i-1 \quad i \quad i+1 \quad N$ p = prob. moving right
Absorbing state (0 or N)

$P_i = P(\text{A wins game} \mid \text{A starts with } \$i)$

$P_i = pP_{i+1} + qP_{i-1}$; $1 \le i \le N-1$; $\begin{cases} P_0 = 0 \\ P_N = 1 \end{cases}$

Difference eq^n: $P_i = x^i$

$\Rightarrow x^i = px^{i+1} + qx^{i-1} \Rightarrow px^2 - x + q = 0 \Rightarrow x = \dfrac{1 \pm \sqrt{1-4pq}}{2 \cdot p}$

$1 - 4pq = 1 - 4p(1-p) = 4p^2 - 4p + 1 = (2p-1)^2 \;\Big|\; x = \dfrac{1 \pm (2p-1)}{2p} \Rightarrow x = 1, q/p$

$P_i = A \cdot x_1^i + B \cdot x_2^i = A + B(q/p)^i$

$P_0 = A + B(q/p)^0 \Rightarrow B = -A$; $1 = A + B(q/p)^N = A - A(q/p)^N$
$\Rightarrow A = \dfrac{1}{1-(q/p)^N}$

$P_i = A + B(q/p)^i = A - A(q/p)^i = A(1-(q/p)^i) = \dfrac{1-(q/p)^i}{1-(q/p)^N}$

Let $y = q/p$; when $p \to q$, $y \to 1$.

$\lim\limits_{y \to 1} \dfrac{1-y^i}{1-y^N} = \lim\limits_{y \to 1} \dfrac{-iy^{i-1}}{-Ny^{N-1}} = i/N$; $\dfrac{N-i}{N}$; $i/N + \dfrac{N-i}{N} = 1$

Let $i = N - i$, $p = 0.49 \mid N = 20$; $P = 0.40 \mid N = 100 \Rightarrow P = 0.02$

What is a random variable? It's a function from the sample space to $\mathbb{R}$. S [H T] $\to$ 1, 0 $\quad P(X=1) = 1/2$, $P(X=0) = 1/2$
event $\{s : X(s) = 1\}$

Def^n (Bernoulli): A r.v. X is said to have Bernoulli distribution if X has two possible values, 0 & 1.
$P(X=1) = p$, $P(X=0) = 1-p = q$ event $\{s : X(s) = 1 \text{ or } 0\}$